

Journal of Geophysical Research

Supporting Information for

**Estimating Magnitude Completeness in Earthquake Catalogs:
A Comparative Study of Catalog-based Methods**

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Introduction

In the present study, we assess nine catalog-based methods, including three newly developed approaches, under a robust evaluation framework specifically tailored for simulating datasets with non-uniform M_c distributions across both spatial and temporal dimensions. The Supporting Information accompanying this study supplements the main article by (i) deriving the probabilistic foundations of the BSReLU transform, (ii) explaining the physical basis of the linear pruning scheme used to impose incompleteness, and (iii) presenting full frequency-magnitude statistics and completeness magnitude results for both synthetic and observed catalogs. It includes an in-depth statistical characterization of both simulating and observed seismicity, along with results of the calibrated M_c^{pred} as a function of M_c^{true} used in generating the simulated datasets with homogeneous M_c , after being iteratively pruned one, two and four times, respectively, as explained in the main text.

Test S1. Probabilistic interpretations of the BSReLU function

The BSReLU transform substitutes the magnitude M in the Gutenberg-Richter (GR) law with a smooth, completeness-aware function that approaches M when $M \gg M_c$, reduces to M_c when $M \ll M_c$, and interpolates continuously between these limits, with its steepness controlled by σ_{M_c} . Below we show that two independent stochastic constructions, a lognormal-exceedance model (Section S1.1) and a geometric Brownian motion (Section S1.2), lead to the identical closed-form expression given in Equation (11).

S1.1 Lognormal-exceedance interpretation

Assume that detection uncertainty is captured by a multiplicative lognormal perturbation,

$$\ln X \sim N(\ln M, \sigma_{M_c}^2) \quad (\text{S1})$$

and define BSReLU as the expected exceedance of X over the completeness threshold M_c :

$$\text{BSReLU}(M) = M_c + E_m[\max(X - M_c, 0)] \quad (\text{S2})$$

where $E_m[\cdot]$ denotes expectation under the above lognormal distribution. This construction replaces the magnitude M in the GR law with the smoothed, completeness-aware term $\text{BSReLU}(M)$. When $M \gg M_c$, almost every realisations of X exceeds M_c , so $E_m[\max(X - M_c, 0)] \approx M - M_c$ and $\text{BSReLU}(M) \approx M$; conversely, when $M \ll M_c$, nearly all realisations fall below M_c , the expectation vanishes, and $\text{BSReLU}(M) \approx M_c$. Thus BSReLU smoothly bridges the incomplete ($M < M_c$) and complete ($M > M_c$) regimes while preserving the GR asymptotes.

Evaluating the expectation in Equation (S2) under the lognormal distribution (S1) yields

$$\text{BSReLU}(M) = M_c + M \Phi\left(\frac{\ln(M/M_c) + \sigma_{M_c}^2/2}{\sigma_{M_c}}\right) - M_c \Phi\left(\frac{\ln(M/M_c) - \sigma_{M_c}^2/2}{\sigma_{M_c}}\right) \quad (\text{S3})$$

where $\Phi[\cdot]$ is the standard normal cumulative distribution function (CDF). Equation (S3) satisfies the desired limits: for $M \gg M_c$ both CDF arguments tend to $+\infty$ and the corresponding $\Phi[\cdot]$'s tend to 1 so that $\text{BSReLU}(M) \rightarrow M$; for $M \ll M_c$ they tend to $-\infty$ and $\text{BSReLU}(M) \rightarrow M_c$. In the limit $\sigma_{M_c} \rightarrow 0$, the expression reduces to the classical cut-off model. Introducing the normalised variable $x = (M - M_c)/\sigma_{M_c}$ and the scaled form $\text{BSReLU}_{\text{scaled}}(x) = [\text{BSReLU}(M) - M_c]/\sigma_{M_c}$ converts Equation (S3) directly into Equation (11) of the main text.

S1.2 Geometric-Brownian-motion interpretation

Alternatively, let $X(t)$ evolve as a geometric Brownian motion (GBM) along the magnitude axis defined for $t = 0$ to $t = 1$,

$$dX(t) = \frac{\sigma_{M_c}^2}{2} X(t) dt + \sigma_{M_c} X(t) dW(t), \quad X(0) = M \quad (S4)$$

where $W(t)$ is a standard Wiener process. The drift $\mu = \sigma_{M_c}^2/2$ is chosen so that the Itô correction cancels, giving

$$\ln X(1) \sim N(\ln M, \sigma_{M_c}^2) \quad (S5)$$

i.e., the end-value at $t = 1$ has exactly the lognormal distribution used in S1.1. Let us define the BSReLU transform as the expected exceedance of the GBM endpoint above M_c :

$$\text{BSReLU}(M) = M_c + E_m[\max\{X(1) - M_c, 0\}] \quad (S6)$$

with the expectation taken over all GBM trajectories satisfying Equation (S4). Because $\ln X(1)$ obeys Equation (S5), the integral in Equation (S6) is identical to that evaluated in S1.1, yielding the same closed form of Equation (S3). Introducing $x = (M - M_c)/\sigma_{M_c}$ and $\text{BSReLU}_{\text{scaled}}(x) = [\text{BSReLU}(M) - M_c]/\sigma_{M_c}$ reproduces Equation (11) in the main text, confirming that the GBM formulation provides an independent yet equivalent probabilistic foundation for BSReLU.

Both BSReLU and AEReLU replace the sharp cut-off in the GR law with a smooth, probabilistic transition. In each case, the transform is defined as an expected exceedance above the completeness threshold M_c : BSReLU is constructed in terms of lognormal fluctuations, yielding a symmetric Gaussian crossover controlled by the width parameter σ_{M_c} ; AEReLU is defined in terms of two different exponential kernels, introducing a single asymmetry parameter β that allows the right-hand convergence to differ from the left-hand one. Both constructions guarantee continuity and differentiability at $x = 0$ and recover the identity function for $M \gg M_c$ and a constant for $M \ll M_c$. Hence, while the mathematical machinery is identical—compute $E_m[\max(X - M_c, 0)]$ under an appropriate noise model—the choice of kernel (Gaussian versus asymmetric exponential) distinguishes BSReLU from AEReLU and yields different flexibility in fitting incomplete catalogs. For brevity we omit the step-by-step derivation of AEReLU; it parallels the BSReLU procedure presented above.

Test S2. Physical rationale of the magnitude pruning function

The seismic wave amplitude (A) generated by an earthquake is approximately related to its magnitude (M) by the relationship $A \sim 10^M$. Given a specific background noise level (L_{noise}), a seismic network can only detect earthquakes whose signal-to-noise ratio (A/L_{noise}) exceeds a certain threshold. Consequently, the probability of earthquake detection increases monotonically with magnitude rather than following an ideal step function. This physical reality justifies modeling the “missing rate” using a continuous function ranging between 0 and 1.

Within a finite magnitude range $[m_{\min}, m_{\max}]$, the common S -shaped (logistic or log-normal) detection probability curves can be approximated at first order by a linear function:

$$P_{\text{miss}}(m) \simeq 1 - \alpha(m - m_{\min}), m_{\min} \leq m \leq m_{\max} \quad (\text{S1})$$

By defining the missing proportion directly as $P_{\text{miss}}(m) = y(m)$, we obtain Equation (14). The linear pruning function thus maintains the physical characteristic of improved detectability with increasing magnitude, while simultaneously being mathematically straightforward and computationally simple. It offers three advantages: (1) Independence from detection models: Equation (14) is not tied to any specific detection probability model under evaluation, thus avoiding the issue of “model correct specification,” where a given model would inherently perform better when fitting data generated by itself. (2) Physical consistency: The missing rate must decrease monotonically with increasing magnitude. The linear function serves as a first-order approximation adequately describing the detectable transition region within a limited magnitude range. (3) Flexibility and control: Parameters c_1 and c_2 are determined by boundary conditions, enabling direct interpretation of $m_{\max}$ as the completeness magnitude (M_c) and facilitating systematic analyses.

A single iteration of pruning can represent a single observational constraint such as sparse station spacing, temporary station outages, or aftershock sequence overlapping. In reality, these constraints often coexist independently. Repeatedly applying the same pruning function N times is equivalent to chaining N independent observational filters. Higher values of N produce steeper curves, representing more severe observational constraints. According to Equation (14), as the number N of observational filters increases, the percentage of missing earthquakes $y(m)$ approaches zero only slowly near $m_{\max}$ (no missing events) but rapidly approaches one near $m_{\min}$ (all events missed). Thus, increased N sharpens the transition from complete missingness to complete detection between $m_{\min}$ and $m_{\max}$, a behavior that closely aligns with real-world seismic monitoring complexities, reflecting a gradual transition from undetectable small earthquakes through partial detectability of intermediate magnitudes to near-certain detection of larger earthquakes.

Figure S3 presents the normalized root-mean-squared error (NRMSE) between observed earthquake catalogs from six study regions and simulated catalogs after

applying N pruning iterations ($N = 1$ to 6). An NRMSE value approaching 0 indicates a closer match between pruned simulation and observational data. Specifically, we first fit the observed earthquake catalogs from the six regions (Figure S2) using the AEReLU function given by Equation (12) to obtain estimates for the Gutenberg-Richter parameters (a and b). These parameters were then incorporated into Equation (14), and following the methodology described in Section 3.1.1, we calculated the NRMSE between each observed catalog and the corresponding simulated catalogs under different pruning scenarios.

Figure S3 shows that each region exhibits a NRMSE closest to 0 at distinct values of $m_{\max}$ (interpreted as the estimated M_c) for different numbers of pruning iterations. For instance, Beijing-Tianjin-Hebei, southeastern coastal, and California have NRMSE closest to 0 at $M = 1.9$, 1.2, and 4.0 respectively after one pruning iteration; Sichuan-Yunnan achieves NRMSE closest to 0 at $M = 1.9$ after two iterations; northern Xinjiang achieves NRMSE closest to 0 at $M = 1.6$ after three iterations; New Zealand reaches NRMSE closest to 0 at $M = 2.6$ after five iterations. These results indicate that different study regions have varying numbers of underlying observational constraints or “filters.” Consequently, we primarily discuss results from three pruning iterations in the main text, while results for one, two, and four iterations are provided in Supporting Information, since we hypothesize that seismic station density, instrumental observation quality, and environmental noise constitute the three primary pruning filters in real-world scenarios.

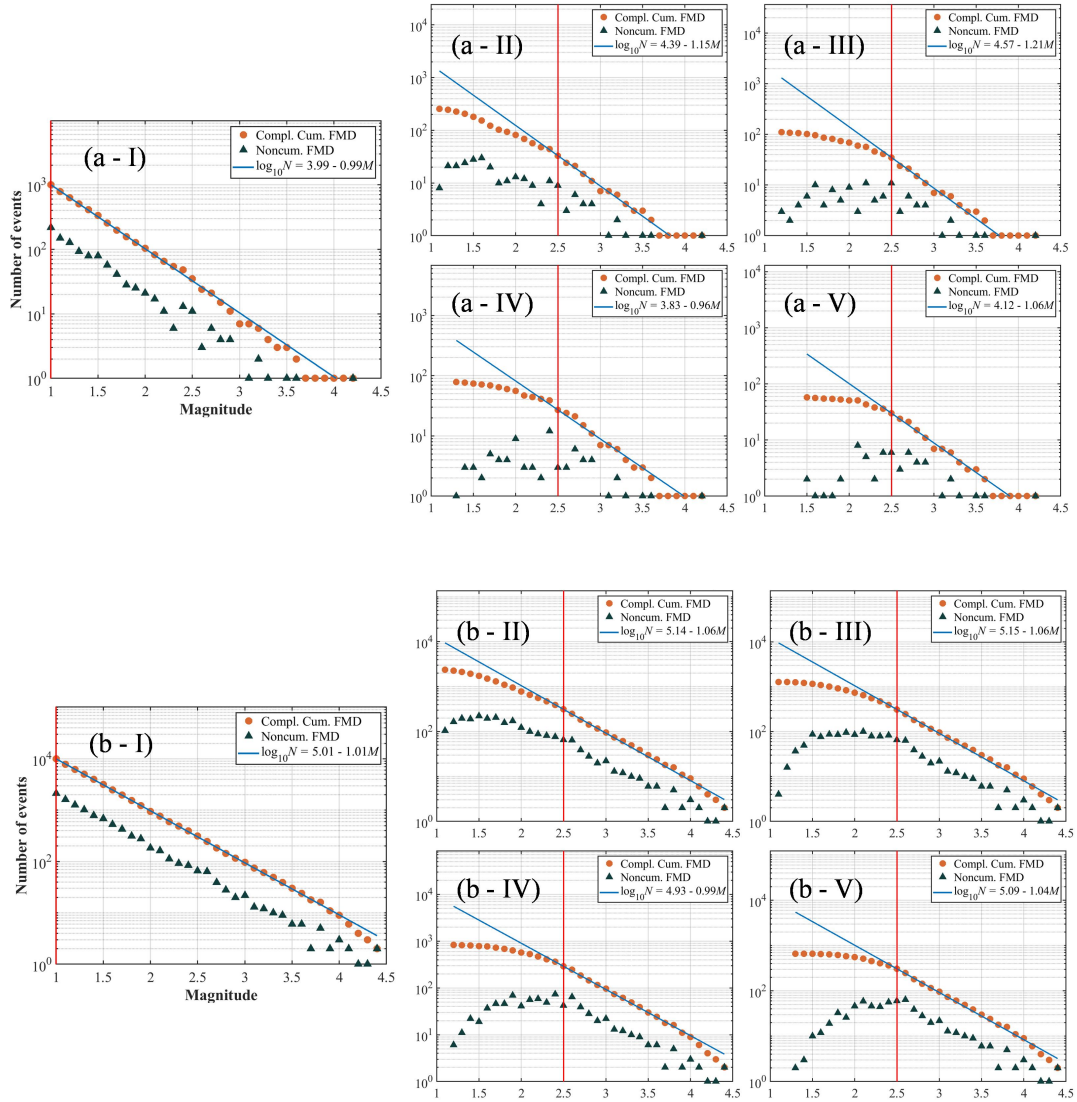


Figure S1. Complementary cumulative distribution (CCFMD) and density distribution (FMD) alongside the Gutenberg-Richter (GR) law, fitted using simulated earthquakes with magnitudes larger than 2.5, for different total number of simulated earthquakes with $M \geq M_{\min} = 1.0$: (a) 10^3 , (b) 10^4 , (c) 10^5 , and (d) 10^6 . For each case, the b -value of the GR law is set to 1.0 to simulate synthetic seismicity, and panel (I) shows the CCFMD and FMD, which follow a pure power law for the entire magnitude range from $M_{\min}$ to $M_{\max}$. Panels (II) to (V) show results from iteratively applying the pruning function to magnitudes from $m_{\min} = 1.0$ to $m_{\max} = 2.5$, one to four times, respectively. By setting M_c^{true} to $m_{\max} = 2.5$, the data are separated into incomplete and power law (complete) parts. In this study, $m_{\max}$ (i.e., M_c^{true}) is varied between 1.0 and 5.0, and each data set, like the one in Figure S1, is generated 200 times. For details on data pruning, refer to the main text.

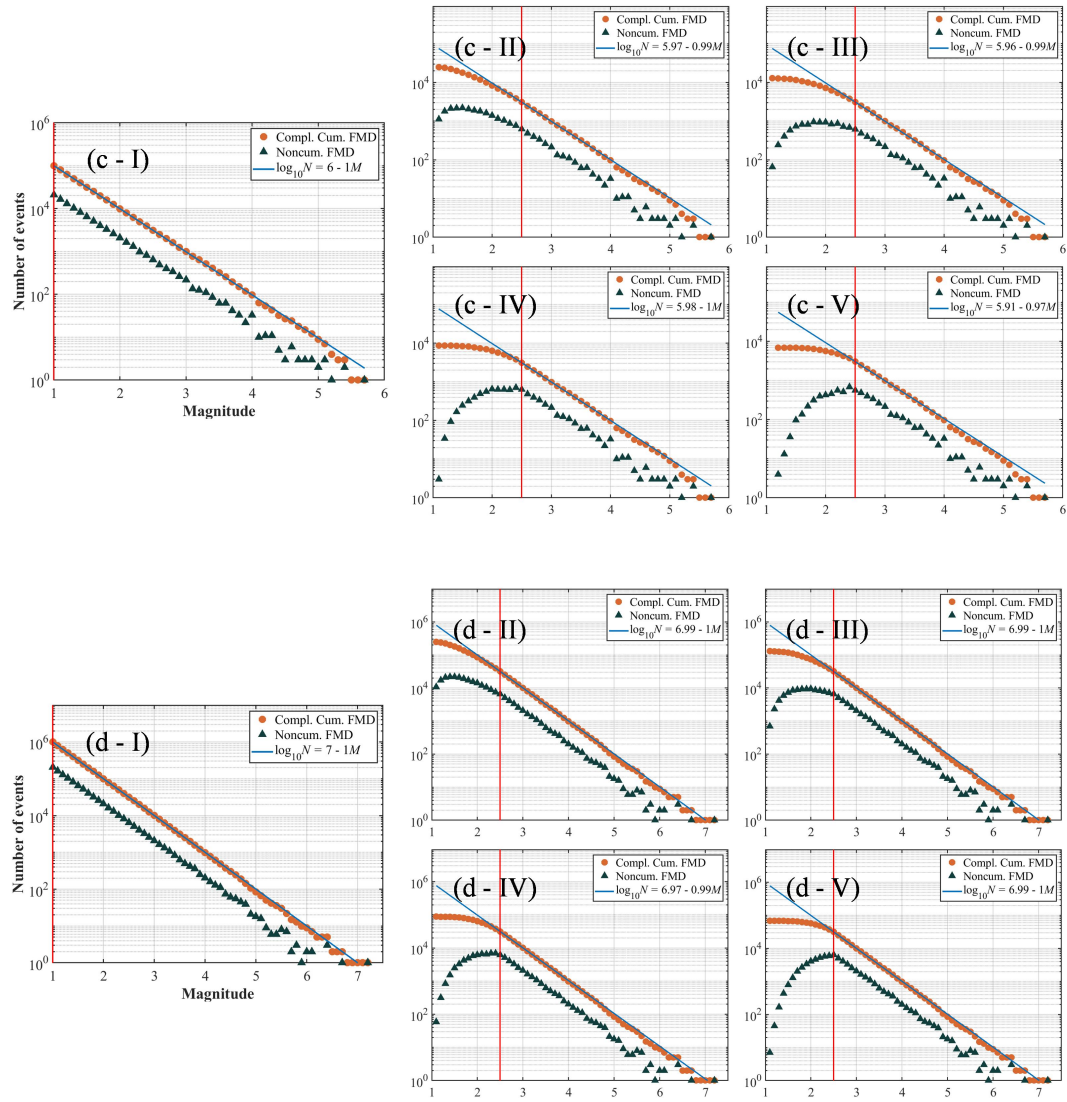


Figure S1. -continued.

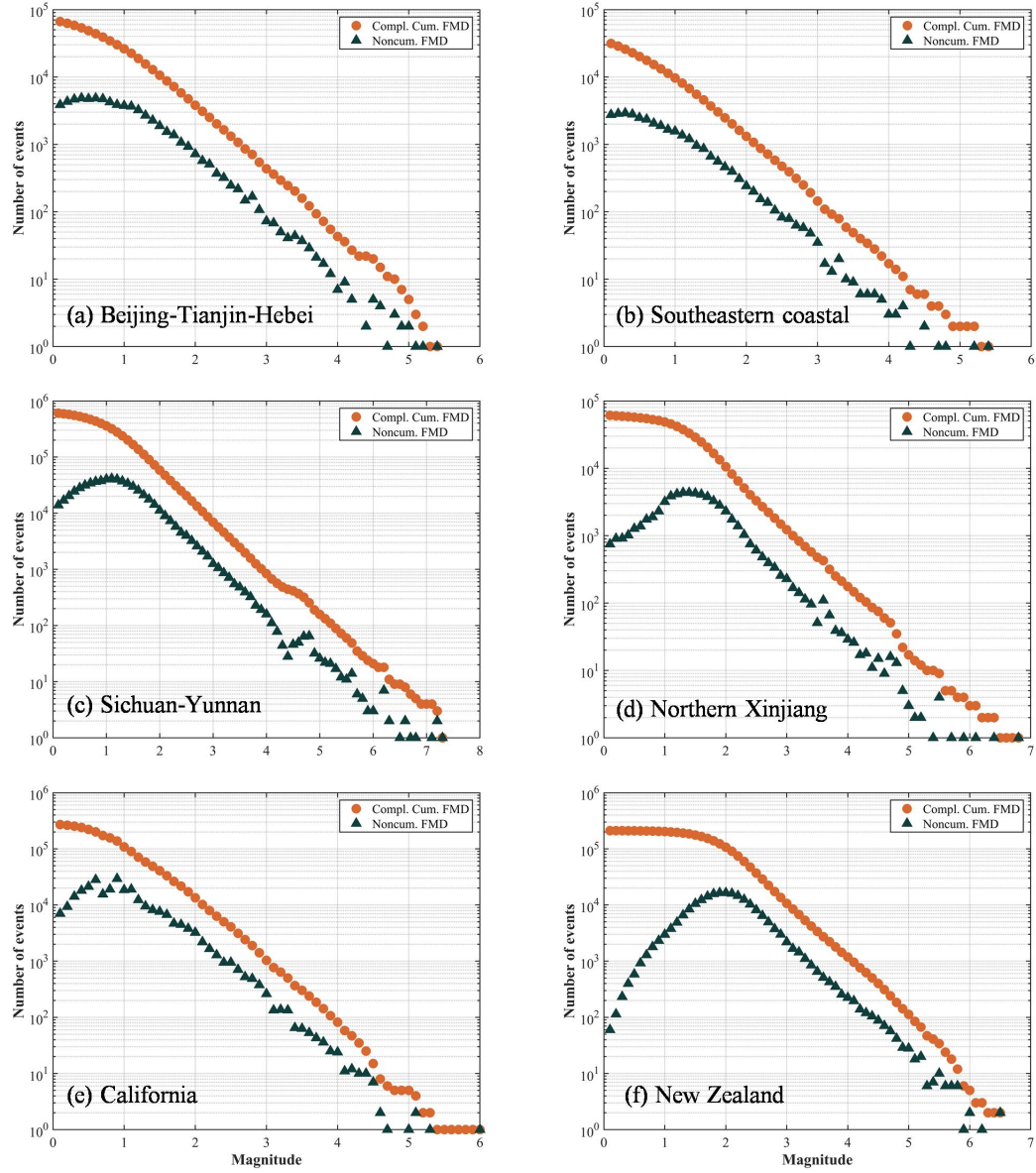


Figure S2. Complementary cumulative distribution (CCFMD) and density distribution (FMD) for the observed seismicity in (a) the Beijing-Tianjin-Hebei region, (b) the southeastern coastal region, (c) the Sichuan-Yunnan region, (d) northern Xinjiang, (e) California, and (f) New Zealand.

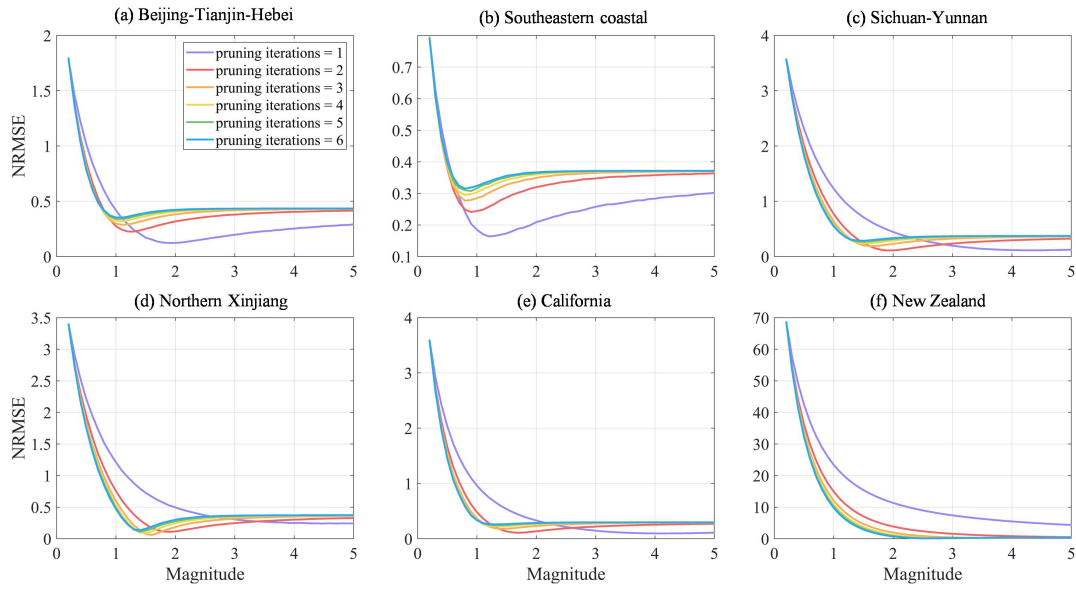


Figure S3. Normalized root-mean-squared error (NRMSE) between the observed catalogs of the six study regions and their synthetic counterparts after N pruning iterations ($N = 1$ to 6). Each observed catalog (see Figure S2) is first fitted with the AEReLU model in Equation (12) to estimate the Gutenberg-Richter parameters (a and b). Using these parameters in Equation (14) and the procedure of Section 3.1.1, we generate pruned synthetic catalogs and compute the NRMSE for every region and pruning level.

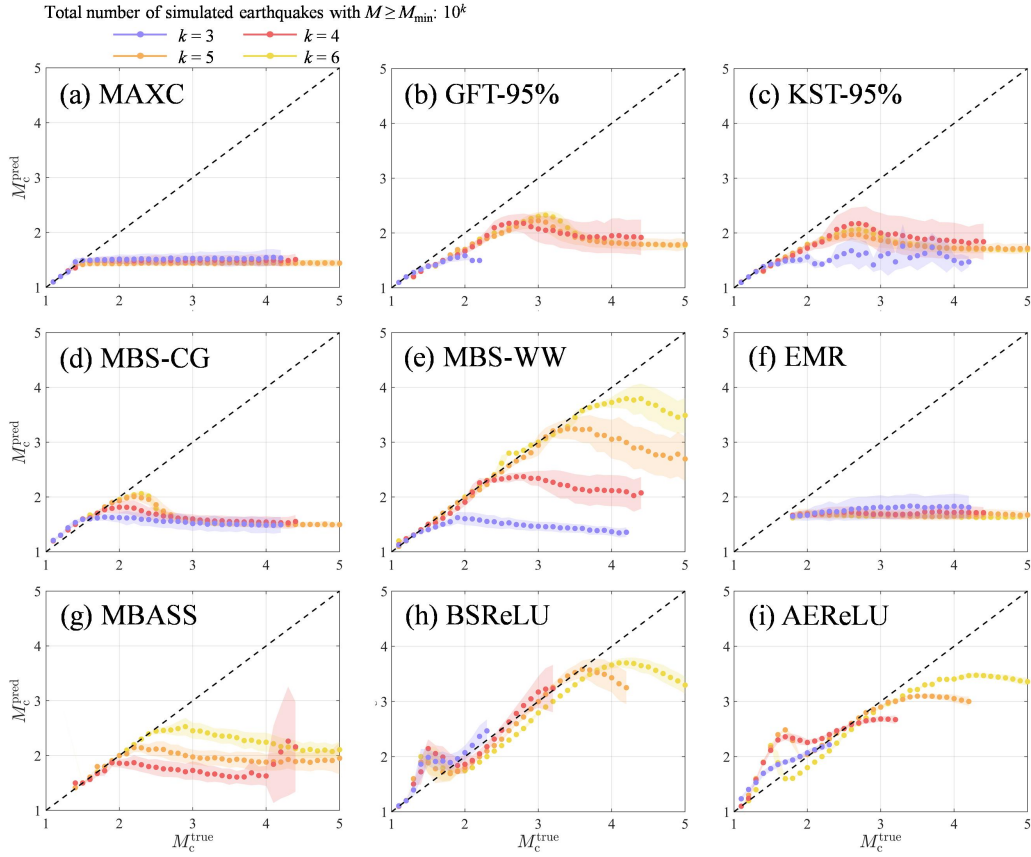


Figure S4. Results of the estimation of the completeness magnitude M_c^{pred} as a function of the true completeness magnitude M_c^{true} for the simulated datasets with homogeneous M_c , after being iteratively pruned one time, as shown in Panels (II) of Figure S1. Other details are the same as those in Figure 2.

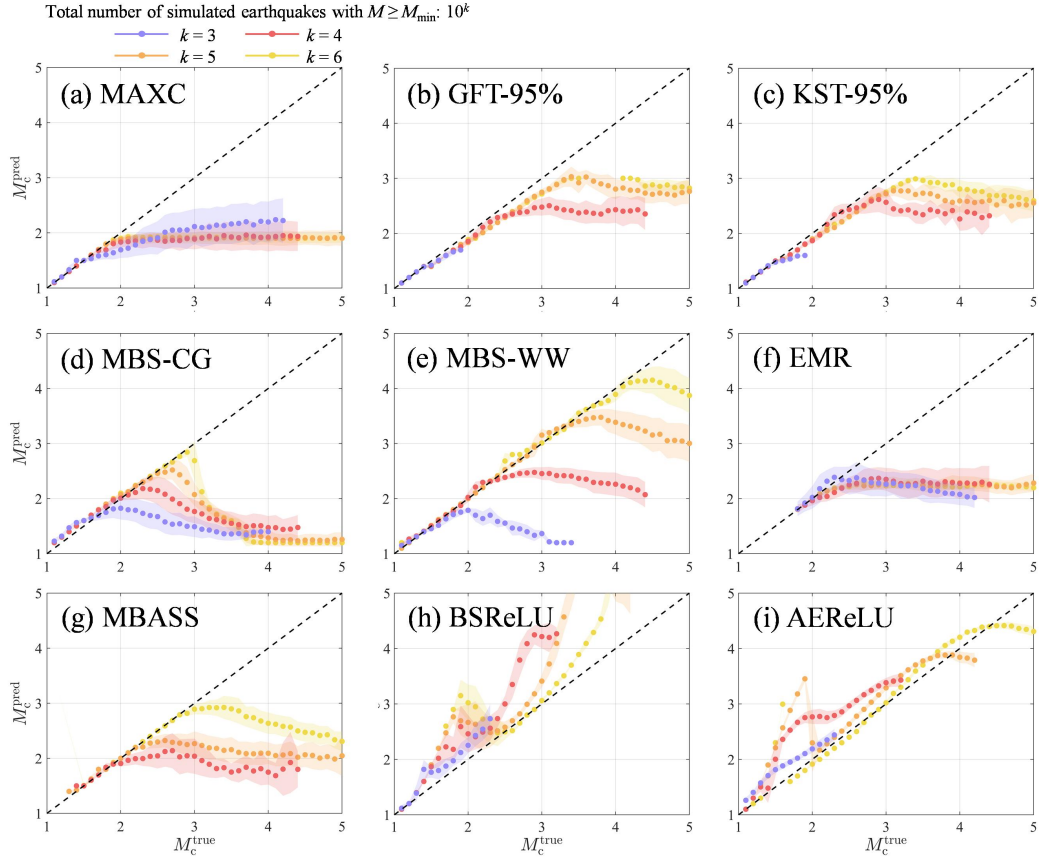


Figure S5. Results of the estimation of M_c^{pred} as a function of M_c^{true} for the simulated datasets with homogeneous M_c , after being iteratively pruned two times, as shown in Panels (III) of Figure S1. Other details are the same as those in Figure 2.

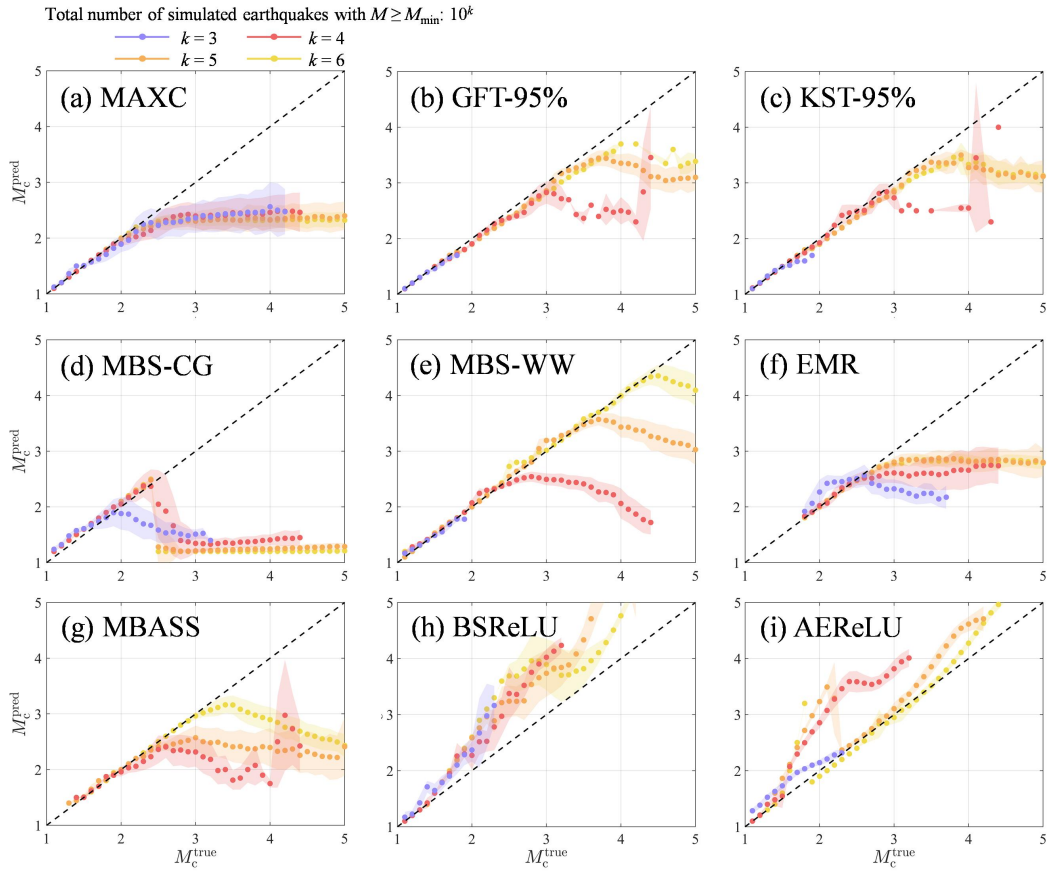


Figure S6. Results of the estimation of M_c^{pred} as a function of M_c^{true} for the simulated datasets with homogeneous M_c , after being iteratively pruned four times, as shown in Panels (V) of Figure S1. Other details are the same as those in Figure 2.